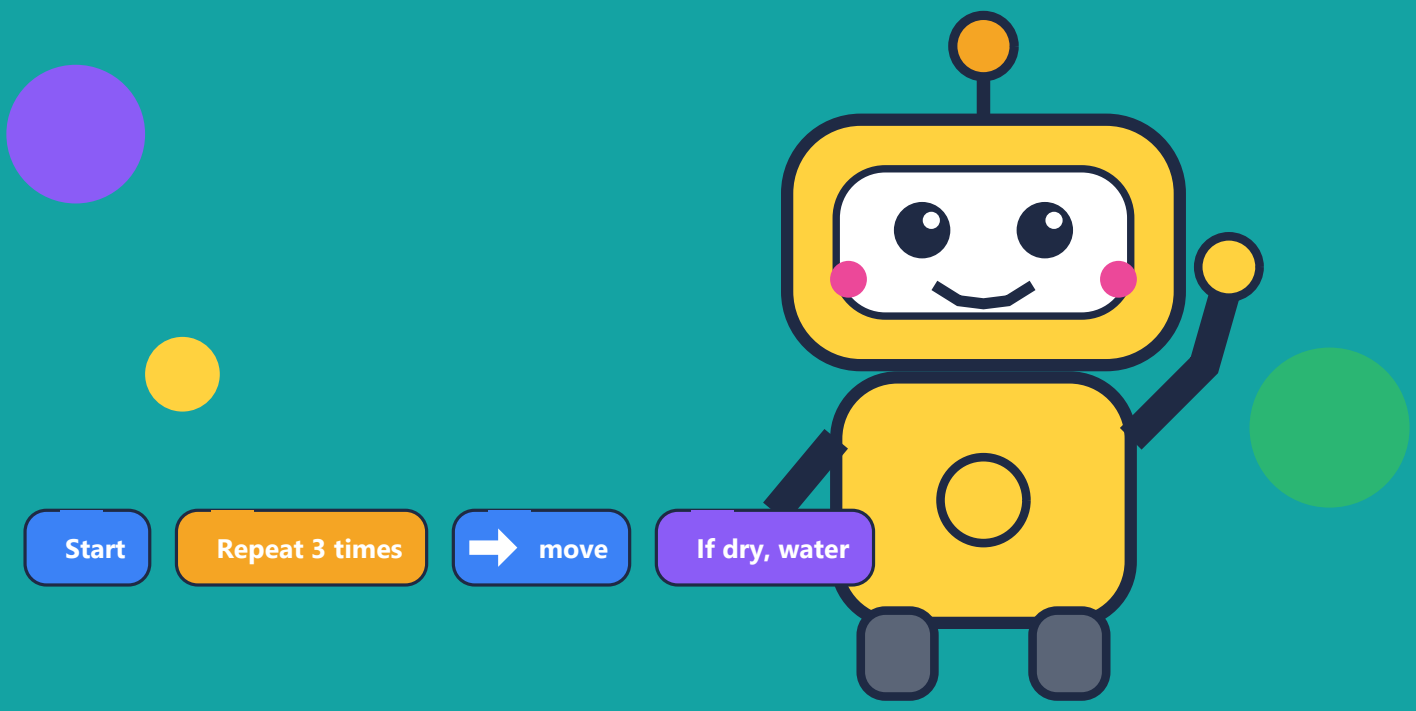
CodeSchool · Class 3

Think Like a Coder

Computational thinking for Class 3

Learn it · See it · Try it · Build it



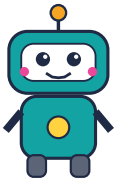
Follows the 14 chapters of NCERT Maths Mela, Class 3

Built for CBSE's Computational Thinking and AI curriculum, 2026-27

Theory, worked examples, practice, coding on paper and challenges

SAMPLE EDITION · DRAFT

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Hello! I am Robo. I think like a coder, and soon you will too! In this book we will solve puzzles, crack secret codes, guide robots and write programs, all on paper.

How this book works

Learn — Robo explains one big idea in simple words, with the new coder words to remember.

Example — A worked example shows the idea step by step.

Try it! — Five practice questions. Circle the answer or write it in the box.

Now you build it — Coding on paper: put code cards in order, fix bugs and write your own programs.

Challenge — A harder puzzle for when you are ready for more.

How did I do? — Colour a face to show how sure you feel.

My code cards

Coders use cards like these. Each colour means something different. You will find cut-out cards at the back of the book.

Step: do this | **Repeat: a loop** | **If: a decision** | **When: an event** | **move**

For teachers and parents

Each chapter goes with the same chapter of the NCERT Class 3 maths book, *Maths Mela*, so it can be used in maths periods. CBSE suggests about 50 hours of computational thinking a year in Classes 3-5. The coding here needs no computer: children act out, write and check programs on paper, which builds the same ideas used later in block coding and AI. Answers are at the back. Encourage children to explain their thinking, not just the answer.

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What's inside

1. What's in a Name? 4
2. Toy Joy 7
3. Double Century 10
4. Vacation with My Nani Maa 13
5. Fun with Shapes 16
6. House of Hundreds - I 19
7. Raksha Bandhan 22
8. Fair Share 25
9. House of Hundreds - II 28
10. Fun at Class Party! 31
11. Filling and Lifting 34
12. Give and Take 37
13. Time Goes On 40
14. The Surajkund Fair 43
Answers 46
Coder words 48
Cut-out code cards 49

Coder skills in this book

Levels: B1 sequences and loops, B2 events, patterns and data, B3 if-then decisions.

Ch	Coder skill	Level	Ch	Coder skill	Level
1	Algorithms and secret codes	B1	8	Rules for sorting and sharing	B2
2	Abstraction: seeing what matters	B1	9	Clues and logical thinking	B2
3	Patterns and loops	B1	10	Sorting: putting things in order	B2
4	Breaking problems into steps	B1	11	If-then rules and comparing	B3
5	Shape patterns and repeat	B2	12	Variables: keeping count	B2
6	Events: when this happens, do that	B2	13	Sequences and loops in time	B2
7	Equal groups and sorting data	B2	14	Grids and robot routes	B1-B2

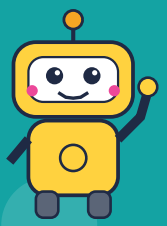
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CHAPTER 1 · LEVEL B1

1 What's in a Name?

Coder skill: Algorithms and secret codes

Goes with NCERT Maths Mela, Class 3, chapter 1

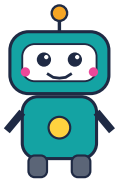


Learn

When you brush your teeth, you follow steps in the same order every day. A list of steps like this is called an **algorithm**.

A **secret code** also follows an algorithm. In a **shift code**, every letter moves forward in the alphabet by the same number. That number is the **key**. Only someone who knows the key can read the message.

algorithm	code	key
steps done in order	a hidden message	the secret number



An algorithm is just a list of steps. Follow them in order and you get the same answer every time!

Example

Code the name RAM with key +1

1. Take the first letter, **R**. Move one letter forward: **S**.
2. Next letter **A** becomes **B**. Then **M** becomes **N**.
3. So **RAM** is coded as **SBN**. To read it back, move each letter one step *back*.

top row: plain letters bottom row: coded (+1)

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A

★ *Coder fact!* Julius Caesar used a shift code more than 2,000 years ago. He moved each letter 3 places forward!

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Try it!

Circle the right answer, or write it in the box.

1 Use key **+1** to code the word **BAT**.

2 Mira coded a word with key **+2**. Her code is **DQQM**. What is the word?

3 Robo coded **SUN** as **TVO**. What key did Robo use?

a) +1 **b)** +2 **c)** +3 **d)** +4

4 Write these names from **shortest to longest**: Meera, Ali, Joseph, Tara.

5 Look at the first letters: **A, C, E, G, ?** Which letter comes next?

a) H **b)** I **c)** J **d)** K

Now you build it

These steps code any word with key +2, but they are jumbled. Number them in the right order.

- Write the new letter down
- Repeat until no letters are left
- Take the next letter of the word
- Move it 2 letters forward

Now use your steps to code your own name with key +2.

Challenge

Robo coded **HELLO** as **KHOOR**. What was the key? Use the same key to decode **FDW**.

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✓ How did I do?

I can follow steps in order.	I can code and decode a word.	I can find the key of a code.
		

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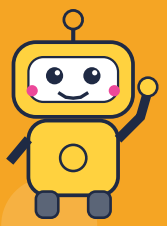
2

CHAPTER 2 · LEVEL B1

Toy Joy

Coder skill: Abstraction: seeing what matters

Goes with NCERT Maths Mela, Class 3, chapter 2

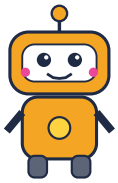


Learn

The same toy looks different from different sides. From the **top**, a drum looks like a circle. From the **side**, it looks like a rectangle.

Computers do this too. A map app shows a city from the top and leaves out details that do not help you find the way. Keeping only the important details is called **abstraction**.

view	abstraction
how a thing looks from one side	keeping only what matters

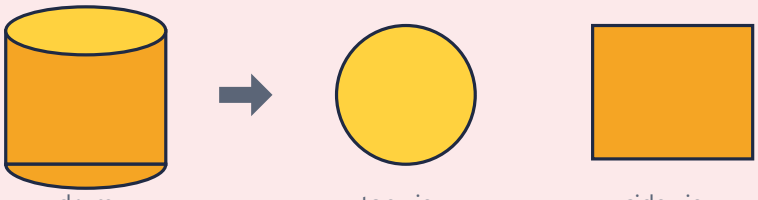


When I look at a map, I only need the roads, not the colour of the houses. Keeping only what matters is called abstraction.

Example

A drum from two sides

1. Look at the drum from the top: you see a **circle**.
2. Look at it from the side: you see a **rectangle**.
3. Both pictures are true. Each shows only some of the details.



drum | top view | side view

★ *Coder fact!* Metro maps leave out most streets and bends, so riders can see the lines and stops clearly. That is abstraction.

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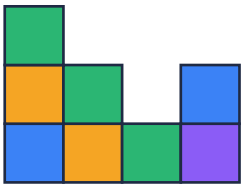
Try it!

Circle the right answer, or write it in the box.

1 Match each toy to its **top view**. Write the letter under each number.



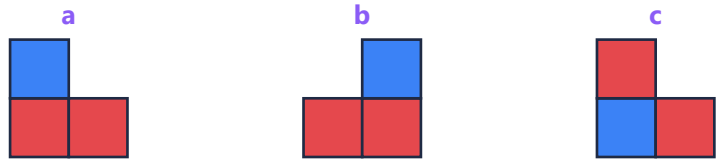
2 How many blocks are in this wall?



3 Robo must reach the park using a map. Which detail matters most?

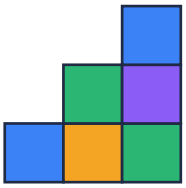
a) colour of houses
b) the roads
c) names of shops
d) the clouds

4 Read the card: *Put 2 red blocks side by side. Put 1 blue block on top of the right one.* Which picture matches?



a) a
b) b
c) c
d) none

5 These steps are built from blocks. How many blocks are needed for the next, taller step (4 blocks high)?



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Now you build it

Write instruction cards so a friend can build this tower without seeing it.



Put a ___ block

on top of / next to

Swap cards with a friend. Do their instructions build the same tower?

Challenge

A tower has 3 layers. The top layer has 1 block, the middle has 4, and the bottom has 9. How many blocks in all? How many would a 4th layer under it have?

How did I do?

I can see a toy from different sides.	I can count blocks in a wall.	I can keep only the details that matter.
☺ 😐 ☹	☺ 😐 ☹	☺ 😐 ☹

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CHAPTER 3 · LEVEL B1

3 Double Century

Coder skill: Patterns and loops

Goes with NCERT Maths Mela, Class 3, chapter 3



Learn

A **loop** repeats the same step again and again. Counting in tens is a loop: start at 20, add 10, add 10, add 10...

A loop needs three things: where to **start**, the **step** to repeat, and **how many times** to repeat it.

loop	pattern
repeat the same step	things that follow a rule

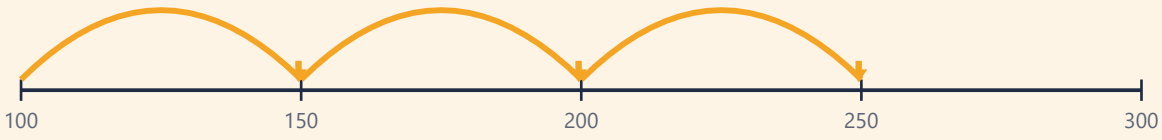


Why write 'add 10' four times when I can say 'repeat 4 times: add 10'? That is a loop!

Example

Start at 100. Repeat 3 times: add 50.

1. Start: **100**.
2. Repeat 1: 100 + 50 = **150**. Repeat 2: **200**. Repeat 3: **250**.
3. The loop stops after 3 repeats. The answer is **250**.



Coder fact! A computer can repeat a loop millions of times in one second, and it never gets bored.

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Try it!

Circle the right answer, or write it in the box.

1 Fill in the next two numbers: **120, 130, 140, ___, ___**

2 What is the loop in **5, 10, 15, 20**?

a) add 2 **b)** add 5 **c)** add 10 **d)** double it

3 Start at 200. Repeat 4 times: add 25. Where do you stop?

4 One number breaks the pattern: **110, 120, 135, 140**. Which one?

a) 110 **b)** 120 **c)** 135 **d)** 140

5 A frog hops 20 each time from 0. How many hops to reach 100?



Now you build it

Make a loop card that goes from 100 to 200. Fill in the blanks.



Can you find two different loops that both reach 200?

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★ Challenge

Robo A starts at 0 and adds 20 each step. Robo B starts at 100 and adds 10 each step. After how many steps do they show the same number? What is it?

✓ How did I do?

I can continue a number pattern.	I can find the step in a loop.	I can write my own loop card.
		

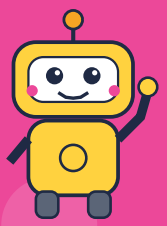
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CHAPTER 4 · LEVEL B1

4 Vacation with My Nani Maa

Coder skill: Breaking problems into steps

Goes with NCERT Maths Mela, Class 3, chapter 4

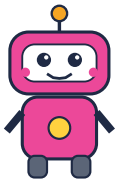


Learn

Going to Nani Maa's house is a big job. We can break it into small steps: pack the bag, go to the bus stop, take the bus, take the train. This is called **decomposition**.

The **order** of steps matters. If a step is in the wrong place, the plan does not work. A mistake in the steps is called a **bug**, and fixing it is called **debugging**.

decompose	bug	debug
break into small steps	a mistake in the steps	find and fix it



A big job feels easy when I break it into small steps. And if something goes wrong, I look for the bug!

Example

The trip to Nani Maa's house

1. Big job: reach Nani Maa's house.
2. Small steps: walk to the bus stop, ride to the station, take the train.
3. Each small step is easy. Done in order, they finish the big job.



★ *Coder fact!* In 1947, engineers found a real moth stuck inside a computer and taped it into their notebook as a 'bug'.

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Try it!

Circle the right answer, or write it in the box.

1 Number these steps for making lemonade (1 to 5): ___ stir it, ___ cut a lemon, ___ pour water in a glass, ___ squeeze the lemon in, ___ add sugar.

2 Find the bug: *1. Put on shoes. 2. Put on socks. 3. Tie the laces.* Which two steps should swap?

a) 1 and 2 **b)** 2 and 3 **c)** 1 and 3 **d)** no bug

3 The train ticket costs Rs 120 and the auto costs Rs 40. Break the cost into parts and find the total.

4 The train leaves at 9 o'clock and the journey takes 3 hours. When does it reach?

a) 10 o'clock **b)** 11 o'clock **c)** 12 o'clock **d)** 1 o'clock

5 Break the job **'get ready for school'** into 4 small steps.

Now you build it

Robo wrote a plan to water the plant, but it has a bug. Find it and fix it.

1. Fill the can with water **2. Pour water on the plant** **3. Walk to the plant**

4. Put the can back

Write the fixed order of the four cards.

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★ Challenge

A recipe has two bugs: *1. Eat the sandwich. 2. Take two slices of bread. 3. Put the cucumber on top. 4. Spread butter.* Write the correct order.

✓ How did I do?

I can break a big job into steps.	I can put steps in order.	I can find and fix a bug.
		

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5

CHAPTER 5 · LEVEL B2

Fun with Shapes

Coder skill: Shape patterns and repeat

Goes with NCERT Maths Mela, Class 3, chapter 5

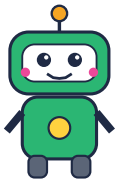


Learn

A **pattern** follows a rule. In a repeating pattern, one group of shapes repeats again and again. This group is called the **core**.

Some patterns **grow**: each step has more than the one before. Spotting patterns is how computers learn to recognise pictures and sounds.

pattern	core	growing
follows a rule	the part that repeats	gets bigger each step



Every pattern has a core: the part that repeats. Find the core and you can continue any pattern!

Example

Find the core

1. Look at the row: triangle, circle, square, triangle, circle, square...
2. The part that repeats is **triangle, circle, square**. That is the core.
3. So the next shape is a **triangle**.



★ *Coder fact!* Some old weaving looms followed patterns punched on cards. Those cards helped inspire the first computers.

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Try it!

Circle the right answer, or write it in the box.

1 Draw the next two shapes.



2 What is the core of this pattern?



a) square, diamond
b) square, square, diamond
c) diamond, square
d) square

3 This pattern grows. How many dots will step 4 have?



4 Which shape breaks the pattern?



a) A
b) C
c) E
d) F

5 An arrow turns: up, right, down, left, ... Which way does it point next?

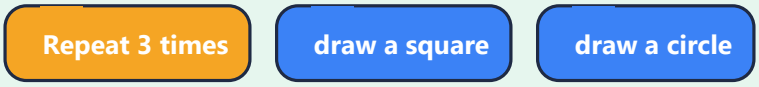


a) up
b) down
c) left
d) right

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Now you build it

Follow this program and draw what it makes in the space below.



Now write your own repeat program for a friend to draw.

Challenge

Step 1 has 2 blocks, step 2 has 4, step 3 has 6. How many blocks will step 10 have?

How did I do?

I can find the core of a pattern.	I can continue a growing pattern.	I can draw from a repeat program.
		

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6

CHAPTER 6 · LEVEL B2

House of Hundreds - I

Coder skill: Events: when this happens, do that

Goes with NCERT Maths Mela, Class 3, chapter 6

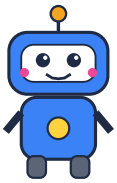


Learn

An **event** is something that happens, like a bell ringing or a button being pressed. A program can say **'when this happens, do that'**.

Numbers are built from **hundreds**, **tens** and **ones**. 254 is 2 hundreds, 5 tens and 4 ones.

event	hundreds, tens, ones
something that happens	the places of a number



When you press my green button, I add 10. When you press my red button, I add 100. Pressing a button is an event!

Example

The counter machine

1. The counter shows **135**. Green button: add 10. Red button: add 100.
2. Press green, green, red: 135 then 145, then 155, then **255**.
3. Each press is an event, and each event runs its own rule.



★ *Coder fact!* When you tap a phone screen, the tap is an event. The app runs the code that belongs to that event.

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Try it!

Circle the right answer, or write it in the box.

1 The counter shows 210. You press green, green. What does it show?

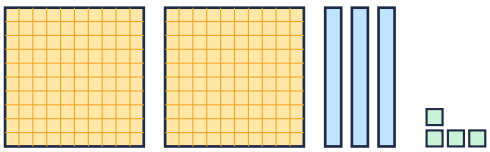
2 The counter shows 405. You press red once. What does it show?

a) 415 **b)** 505 **c)** 406 **d)** 605

3 Which presses take the counter from 120 to 340?

a) red, green
b) red, red, green, green
c) green x 22
d) red, red

4 What number do these blocks show?



5 Use the digits 3, 0 and 7 once each. What is the **largest** number you can make?

Now you build it

Design your own counter machine. Choose what each button does.

When yellow is pressed | add ___ | When blue is pressed | take away ___

Start at 500. Ask a friend to press 3 buttons and tell you the number.

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★ Challenge

The counter starts at 0. What is the smallest number of presses to reach 320?

✓ How did I do?

I can follow 'when this happens' rules.	I can read hundreds, tens and ones.	I can make the largest number.
		

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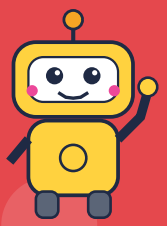
CHAPTER 7 · LEVEL B2

7

Raksha Bandhan

Coder skill: Equal groups and sorting data

Goes with NCERT Maths Mela, Class 3, chapter 7

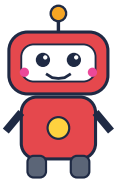


Learn

When we make **equal groups**, we repeat the same step. 'Repeat 3 times: put 2 sweets' gives 6 sweets.

We can also **sort** things into groups by a rule, like colour. Then we count each group and show it in a table or picture. This is called **data**.

equal groups	data	sort
same number in each	facts we collect and count	group by a rule

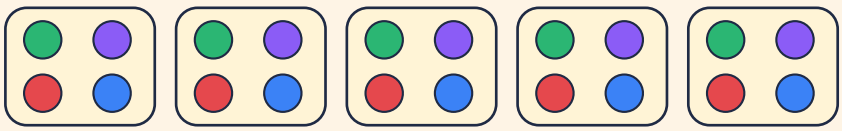


Putting 4 rakhis in each of 5 packets is a loop: repeat 5 times, put 4 rakhis. That makes 20!

Example

Rakhis in packets

1. There are 5 packets with 4 rakhis in each.
2. Loop: repeat 5 times, add 4: 4, 8, 12, 16, **20**.
3. So there are **20 rakhis** in all.



★ *Coder fact!* Shops count what they sell every day. This data tells them what to stock more of next time.

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Try it!

Circle the right answer, or write it in the box.

1 There are 3 boxes with 6 laddoos in each. How many laddoos in all?

2 Sort these rakhis by colour. How many are **red**?



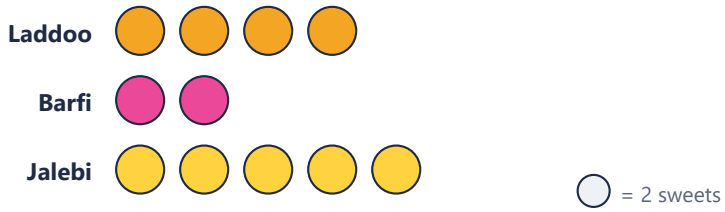
a) 2 **b)** 3 **c)** 4 **d)** 5

3 18 sweets are shared equally into 3 boxes. How many sweets in each box?

4 Which loop makes **15**?

a) repeat 4 times: add 4
b) repeat 3 times: add 5
c) repeat 5 times: add 2
d) repeat 2 times: add 6

5 Read the picture graph. How many jalebis were made? (Each circle = 2 sweets.)



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Now you build it

Write a loop card for 6 friends who each get 2 rakhis.

Repeat ___ times **give ___ rakhis**

How many rakhis do you need? Show it with a drawing.

Challenge

Riya has 3 brothers. Each brother gets 2 rakhis and 4 sweets. How many things does Riya give in all?

How did I do?

I can count equal groups with a loop.	I can sort and count data.	I can read a picture graph.
		

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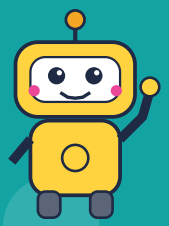
8

CHAPTER 8 · LEVEL B2

Fair Share

Coder skill: Rules for sorting and sharing

Goes with NCERT Maths Mela, Class 3, chapter 8

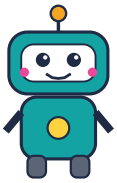


Learn

To share fairly, everyone gets the **same amount**. Cutting something into 2 equal parts gives two **halves**.

A **rule** tells us what to do with each thing. An **if-else** rule has two choices: *if* something is true, do one thing; *else*, do the other.

half	fair	if-else
one of 2 equal parts	everyone gets the same	a rule with two choices

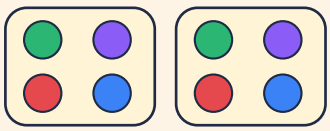


If it is red, it goes in box A. Else it goes in box B. My if-else rule never makes a mistake!

Example

Sharing 8 laddoos between 2 friends

1. Rule: give one to you, one to me. **Repeat** until none are left.
2. After 4 repeats, each friend has **4 laddoos**.
3. Each friend has half of the laddoos. The share is fair.



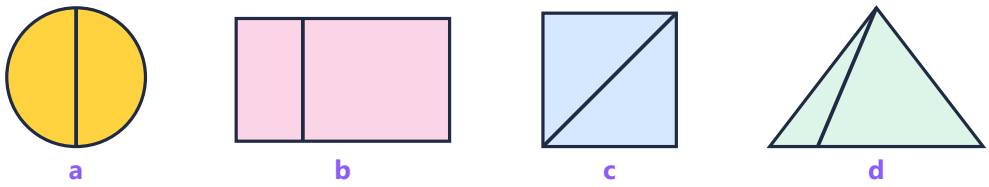
★ *Coder fact!* A traffic light follows a rule: if the light is red, cars stop; else they may go.

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Try it!

Circle the right answer, or write it in the box.

1 Which picture shows **two equal halves**? (There are two.)



a b c d

a) a and c **b)** a and b **c)** b and d **d)** c and d

2 Rule: *if the number is more than 50, put it in BIG, else put it in SMALL*. Where does **50** go?

a) BIG **b)** SMALL

3 Sort into EVEN and ODD: 12, 7, 20, 15, 4.

4 12 pencils are shared fairly among 4 children. How many does each child get?

5 Is the rule '*one for you, two for me*' fair?

a) yes **b)** no

Now you build it

Write an if-else rule to sort fruits into two baskets.

If the fruit is ___ | put it in basket A | else | put it in basket B

Test your rule with apple, banana and mango. Does every fruit go somewhere?

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★ Challenge

3 friends share 7 cookies fairly. How many does each get? How many are left over? What could they do with the leftovers?

✓ How did I do?

I can tell if a share is fair.	I can find halves.	I can use an if-else rule.
		

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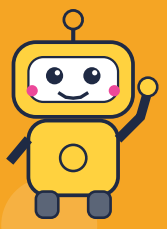
9

CHAPTER 9 · LEVEL B2

House of Hundreds - II

Coder skill: Clues and logical thinking

Goes with NCERT Maths Mela, Class 3, chapter 9

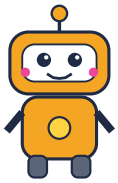


Learn

A detective uses **clues**. Each clue rules out some answers. When only one answer fits every clue, the mystery is solved.

Sometimes a clue tells us nothing new. Good detectives notice which clues are **enough** and which are **extra**.

clue	logic
a hint that rules some answers out	thinking step by step to be sure



Each clue crosses out some numbers. Keep going until only one number is left: that is the answer!

Example

Find the mystery number

1. Clue 1: it is between 300 and 400, so it starts with **3**.
2. Clue 2: the tens digit is 2: **32**_. Clue 3: the ones digit is 5.
3. The mystery number is **325**.

★ *Coder fact!* Detectives and doctors both use clues to rule out answers, step by step, just like you.

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Try it!

Circle the right answer, or write it in the box.

1. I am between 500 and 600. My tens digit is 4 and my ones digit is 0. Who am I?

2. I am a 3-digit number. All my digits are the same and they add up to 15. Who am I?

3. I am more than 230 and less than 240. I am an even number and my ones digit is 8. Who am I?

4. Clues: *A: it is between 100 and 200. B: its tens digit is 6. C: its ones digit is 3. D: it is less than 500.* Which clue is extra?

   a) A **b)** B **c)** C **d)** D

5. Cross out the numbers that are **not** odd. Which numbers are left: 410, 413, 417, 420, 425?

Now you build it

Think of a 3-digit number. Write 3 clues so a friend can find it.

Clue 1: It is between ___ and ___

Clue 2: My tens digit is ___

Clue 3: My ones digit is ___

Can your friend solve it with only two of your clues?

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★ Challenge

I am between 200 and 300. My tens digit and ones digit are the same, and all my digits add up to 10. Who am I?

✓ How did I do?

I can use clues to find a number.	I can spot an extra clue.	I can write my own clues.
		

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CHAPTER 10 · LEVEL B2

10

Fun at Class Party!

Coder skill: Sorting: putting things in order

Goes with NCERT Maths Mela, Class 3, chapter 10



Learn

Sorting means putting things in order, like shortest to tallest. Computers sort lists of names, prices and marks all the time.

One way to sort: **compare** two neighbours, **swap** them if the bigger one is first, then move one place to the right. Repeat until no swaps are needed.

sort	compare	swap
put in order	see which is bigger	change places



To sort, I compare two neighbours. If they are in the wrong order, I swap them. Then I move on and repeat!

Example

Sort 5, 2, 4 from smallest to biggest

1. Compare 5 and 2: 5 is bigger and comes first, so **swap**: 2, 5, 4.
2. Compare 5 and 4: **swap**: 2, 4, 5.
3. Check again: no more swaps needed. Sorted: **2, 4, 5**.



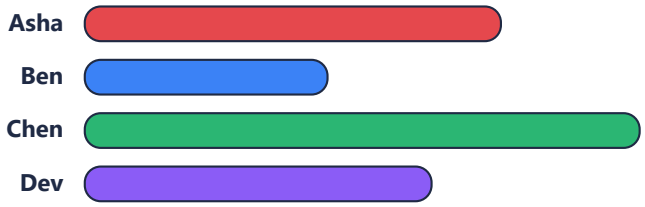
★ *Coder fact!* The contact list on a phone is kept in A to Z order by a sorting algorithm.

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Try it!

Circle the right answer, or write it in the box.

1 Write the children from **shortest ribbon to longest**.



2 Do one round of compare-and-swap on **4, 2, 5, 1**. What is the list after the round?

a) 2, 4, 1, 5 **b)** 1, 2, 4, 5 **c)** 2, 4, 5, 1 **d)** 4, 2, 1, 5

3 Which tool would you use to measure the length of a party table?

a) clock **b)** measuring tape **c)** weighing scale **d)** jug

4 Arrange these cups from **least to most** juice: 300 ml, 150 ml, 500 ml, 250 ml.

5 In a line sorted by height, Tina is 3rd from the front and there are 7 children. How many are behind her?

Now you build it

Put these sorting cards in the right order to make a sorting program.

Move one place right | **Compare two neighbours** | **Repeat until no swaps** | **If the first is bigger, swap**

Use your program to sort the numbers on 4 cards: 9, 3, 7, 1.

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★ Challenge

How many swaps does it take to sort **3, 2, 1** into **1, 2, 3** using compare-and-swap?

✓ How did I do?

I can put things in order.	I can compare and swap.	I can write a sorting program.
		

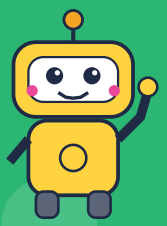
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CHAPTER 11 · LEVEL B3

11 Filling and Lifting

Coder skill: If-then rules and comparing

Goes with NCERT Maths Mela, Class 3, chapter 11

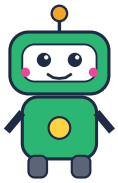


Learn

An **if-then** rule makes a decision: *if* something is true, *then* do this. If it is raining, then take an umbrella.

A **balance** helps us compare weights. The heavier side goes **down**. If both sides are level, they weigh the same.

if-then	heavier	capacity
a rule that makes a decision	weighs more	how much a thing can hold



If the jug is full, then I stop pouring. If-then rules help me decide what to do next!

Example

The tile rule

1. Rule: *if you are on a white tile, then hop 1. If you are on a grey tile, then hop 2.*
2. Start on tile 1 (white): hop to 2. Tile 2 is grey: hop to 4. Tile 4 is white: hop to 5.
3. Each tile tells Robo what to do next.



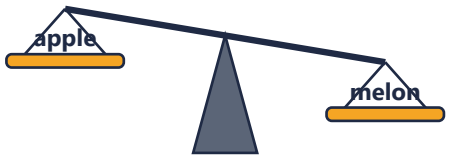
★ *Coder fact!* Washing machines use if-then rules: if the drum is full of water, then stop filling.

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Try it!

Circle the right answer, or write it in the box.

1 Which is heavier?



a) apple **b)** melon **c)** same

2 Use the tile rule from the example on this strip. Start on tile 1. Which tiles does Robo land on?



3 Rule: *if it is sunny, then wear a cap, else take an umbrella*. What do you take on a rainy day?

a) cap **b)** umbrella **c)** both **d)** nothing

4 A bucket fills 8 mugs. A jug fills 3 mugs. Which holds more?

a) bucket **b)** jug **c)** same

5 The bag is heavier than the book. The book is heavier than the pencil. Which is the lightest?

a) bag **b)** book **c)** pencil

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Now you build it

Write if-then rules for watering a plant.

If the soil is dry | then pour one mug | If the soil is wet | then ___

What should happen if the pot is already full of water?

Challenge

You have 4 bags and a balance. What is the smallest number of weighings to be sure which bag is the heaviest?

How did I do?

I can follow an if-then rule.	I can compare weights on a balance.	I can compare how much things hold.
		

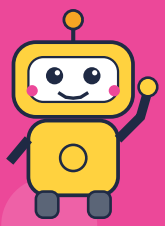
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CHAPTER 12 · LEVEL B2

12 Give and Take

Coder skill: Variables: keeping count

Goes with NCERT Maths Mela, Class 3, chapter 12

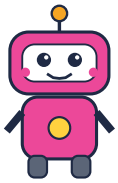


Learn

A **variable** is like a labelled box that holds a number. The number can change when we give or take.

To follow the changes, we **trace**: we write down the number in the box after every step. Programmers do this to check their programs.

variable	trace
a named box for a number	write the number after each step



A variable is a box with a name. My 'marbles' box holds a number, and the number can change!

Example

Trace Sonu's marbles

1. Start: marbles = **20**.
2. Sonu wins 5: marbles = 25. Sonu gives 8 to Rani: marbles = **17**.
3. The box called 'marbles' changed after every step.

Step	What happens	marbles
Start		20
1	wins 5	25
2	gives 8	17

★ *Coder fact!* In a video game, your score is a variable. It changes every time you win or lose points.

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Try it!

Circle the right answer, or write it in the box.

1 Start: coins = 30. Get 10, give 5, get 2. How many coins at the end?

2 Fill the trace table: start 50 stickers, give 15, get 20.

Step	What happens	stickers
Start		50
1	give 15	
2	get 20	

3 You pay Rs 100 for a toy that costs Rs 65. How much change do you get?

4 After getting 12 more beads, Mona has 50 beads. How many did she have at the start?

a) 38 **b)** 42 **c)** 62 **d)** 48

5 Ravi has 15 cards and Gita has 9. Ravi gives Gita 3. Who has more now?

a) Ravi **b)** Gita **c)** same

Now you build it

Write a program for a game score. Use a variable called **score**.

Set score to 0 | When you catch a ball | add ___ to score | When you drop a ball | take ___ from score

Play: catch, catch, drop, catch. Trace the score.

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★ Challenge

Asha and Bala each start with 10 toffees. Asha gives Bala 2, then Bala gives Asha 5. How many does each have?

✓ How did I do?

I can use a variable to keep count.	I can trace a number step by step.	I can find the start by undoing steps.
		

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13

CHAPTER 13 · LEVEL B2

Time Goes On

Coder skill: Sequences and loops in time

Goes with NCERT Maths Mela, Class 3, chapter 13



Learn

Time follows a **sequence**: morning, afternoon, evening, night. Our daily routine is a sequence of steps.

The days of the week make a **loop** of 7. The clock hands go round in loops too. Loops help us work out what day it will be.

sequence	loop
things in order	comes back to the start



Days of the week go round in a loop of 7. After Sunday comes Monday again!

Example

What day is it 10 days after Monday?

1. 7 days after Monday is Monday again (one full loop).
2. 3 more days: Tuesday, Wednesday, **Thursday**.
3. So 10 days after Monday is **Thursday**.

Mon	Tue	Wed	Thu	Fri	Sat	Sun

Coder fact! Computers keep time with a tiny clock inside that ticks millions of times every second.

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Try it!

Circle the right answer, or write it in the box.

1 Number these in order (1 to 4): ___ lunch, ___ wake up, ___ dinner, ___ go to school.

2 Today is Friday. What day will it be in 3 days?

a) Sunday **b)** Monday **c)** Tuesday **d)** Saturday

3 In a month, Sundays fall on the 3rd, 10th and 17th. What is the date of the next Sunday?

4 What time does each clock show?



5 Look at the timeline. What happens just before lunch?



a) school **b)** games **c)** home

Now you build it

Write a loop that says the days of the week.

Start at Monday **Repeat ___ times** **say the day, then go to the next day**

How many times must the loop repeat to say every day once?

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★ Challenge

Neha's birthday is 15 days after Wednesday. On what day is her birthday?

✓ How did I do?

I can put events in time order.	I can use the week loop.	I can read a clock.
		

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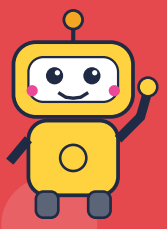
14

CHAPTER 14 · LEVEL B1-B2

The Surajkund Fair

Coder skill: Grids and robot routes

Goes with NCERT Maths Mela, Class 3, chapter 14

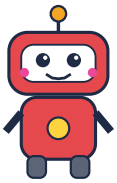


Learn

A **grid** has rows and columns. We can name a square with a letter and a number, like **C2**.

A **program** is a set of instructions for a computer or a robot. Robo moves one square for each arrow. A program that hits a wall has a bug!

grid	program	route
rows and columns of squares	instructions a robot follows	the path you take

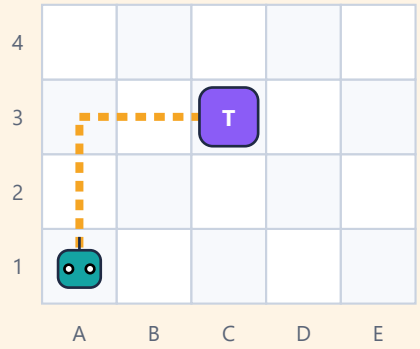


Give me arrow cards and I will follow them exactly. One arrow = one square. That is my program!

Example

Take Robo to the toy stall

1. Robo is at **A1**. The toy stall (T) is at **C3**.
2. Program: up, up, right, right.
3. Robo moves 4 squares and reaches the stall. Another route works too: right, right, up, up.



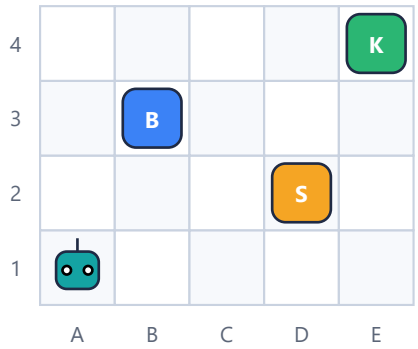
★ *Coder fact!* Robots in big warehouses move around grids, following programs, to fetch parcels.

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Try it!

Circle the right answer, or write it in the box.

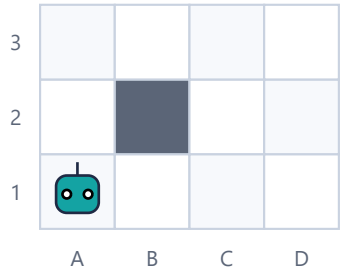
1 Robo starts at A1 and follows: **right, right, up, right**. Which stall does Robo reach?



a) S (sweets) **b)** B (bangles) **c)** K (kites)

2 Write the **shortest** program to take Robo from A1 to the kite stall K at E4 in the grid above.

3 Robo is at A1 and the grey square is a wall. Program: **up, right, right**. Where is the bug?



a) the 1st arrow **b)** the 2nd arrow **c)** the 3rd arrow **d)** no bug

4 Make this program shorter with a loop: **right, right, right, right**.

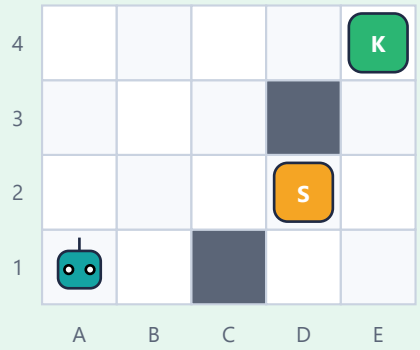
5 Which square is 2 to the right of B1 and 1 up?

a) D2 **b)** C2 **c)** D1 **d)** B3

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Now you build it

Plan Robo's trip around the fair. Start at A1, visit the sweet stall S, then the kite stall K, without crossing the grey walls.



up down left right

Draw your arrows in order. Count them. Can a friend find a shorter route?

Challenge

Write a program with a loop that takes Robo from A1 to D4 along the diagonal staircase: right, up, right, up, right, up.

How did I do?

I can name squares on a grid.	I can follow and write a robot program.	I can find a bug in a route.

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Answers

Many questions have more than one good answer. Accept any answer that follows the rule and can be explained.

1. What's in a Name?

1. CBU
2. BOOK
3. a) +1
4. Ali, Tara, Meera, Joseph
5. b) I
Build it: Take the next letter; Move it 2 letters forward; Write the new letter down; Repeat until no letters are left
Challenge: Key +3; FDW = CAT

2. Toy Joy

1. 1 ball: B; 2 dice: A; 3 party cap: C
2. 8
3. b) the roads
4. b
5. 10 blocks (1 + 2 + 3 + 4)
Build it: Example: Put a blue block. Put an orange block on top of it. Put an orange block next to the blue block.
Challenge: 14 blocks; the 4th layer has 16 (4 x 4)

3. Double Century

1. 150, 160
2. b) add 5
3. 300
4. c) 135
5. 5 hops
Build it: Any loop that adds up to 100: e.g. start 100, repeat 2 times add 50; or repeat 4 times add 25; or repeat 10 times add 10.
Challenge: After 10 steps both show 200: a double century!

4. Vacation with My Nani Maa

1. Example order: 1 cut a lemon, 2 pour water in a glass, 3 squeeze the lemon in, 4 add sugar, 5 stir it
2. a) 1 and 2
3. Rs 160
4. c) 12 o'clock
5. Example: wake up, brush teeth, bathe and dress, pack the school bag
Build it: Fill the can; walk to the plant; pour water on the plant; put the can back (steps 2 and 3 were swapped)
Challenge: 2. Take two slices of bread; 4. Spread butter; 3. Put the cucumber on top; 1. Eat the sandwich

5. Fun with Shapes

1. star, circle
2. b) square, square, diamond
3. 7
4. c) E
5. a) up
Build it: square, circle, square, circle, square, circle
Challenge: 20 (add 2 each step)

6. House of Hundreds - I

1. 230
2. b) 505
3. b) red, red, green, green
4. 234
5. 730
Build it: Open answer
Challenge: 5 presses: red, red, red, green, green

7. Raksha Bandhan

1. 18
2. c) 4
3. 6
4. b) repeat 3 times: add 5
5. 10
Build it: Repeat 6 times: give 2 rakhis = 12 rakhis
Challenge: 18 (6 rakhis + 12 sweets)

8. Fair Share

1. a) a and c
2. SMALL (50 is not more than 50)
3. Even: 12, 20, 4. Odd: 7, 15
4. 3
5. no: one child gets twice as many
Build it: Example: If the fruit is yellow, basket A; else basket B
Challenge: 2 each, 1 left over; they could break it into 3 equal pieces

9. House of Hundreds - II

1. 540
2. 555
3. 238
4. d) D, because clue A already says it is less than 200; the number is 163
5. 413, 417, 425
Build it: Open answer
Challenge: 244

10. Fun at Class Party!

1. Ben, Dev, Asha, Chen
2. a) 2, 4, 1, 5
3. b) measuring tape
4. 150 ml, 250 ml, 300 ml, 500 ml
5. 4
Build it: Compare two neighbours; If the first is bigger, swap; Move one place right; Repeat until no swaps. 9, 3, 7, 1 becomes 1, 3, 7, 9
Challenge: 3 swaps: (3,2), (3,1), (2,1)

11. Filling and Lifting

1. b) melon
2. 1, 2, 4, 6, 7 then hop 2 to the end
3. b) umbrella
4. a) bucket
5. c) pencil
Build it: Example: If the soil is wet, then do not water
Challenge: 3: compare bags 1 and 2, then the heavier with bag 3, then the heavier with bag 4

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12. Give and Take

1. 37
2. 35, then 55
3. Rs 35
4. a) 38
5. c) same: 12 each
Build it: Example (add 2, take 1): 2, 4, 3, 5
Challenge: Asha 13, Bala 7

13. Time Goes On

1. 1 wake up, 2 go to school, 3 lunch, 4 dinner
2. b) Monday
3. 24th
4. A 3:00, B 6:30, C 9:00
5. b) games
Build it: Repeat 7 times
Challenge: Thursday (14 days = 2 loops, then 1 more day)

14. The Surajkund Fair

1. a) S
2. 7 arrows: 4 right and 3 up, in any order
3. b) the 2nd arrow hits the wall at B2
4. Repeat 4 times: right
5. a) D2
Build it: Example: up, right, right, right (S), right, up, up (K)
Challenge: Repeat 3 times: right, up

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Coder words

abstraction	keeping only what matters	Ch 2
algorithm	steps done in order	Ch 1
bug	a mistake in the steps	Ch 4
capacity	how much a thing can hold	Ch 11
clue	a hint that rules some answers out	Ch 9
code	a hidden message	Ch 1
compare	see which is bigger	Ch 10
core	the part that repeats	Ch 5
data	facts we collect and count	Ch 7
debug	find and fix it	Ch 4
decompose	break into small steps	Ch 4
equal groups	same number in each	Ch 7
event	something that happens	Ch 6
fair	everyone gets the same	Ch 8
grid	rows and columns of squares	Ch 14
growing	gets bigger each step	Ch 5
half	one of 2 equal parts	Ch 8
heavier	weighs more	Ch 11
hundreds, tens, ones	the places of a number	Ch 6
if-else	a rule with two choices	Ch 8
if-then	a rule that makes a decision	Ch 11
key	the secret number	Ch 1
logic	thinking step by step to be sure	Ch 9
loop	repeat the same step	Ch 3
pattern	things that follow a rule	Ch 3
program	instructions a robot follows	Ch 14
route	the path you take	Ch 14
sequence	things in order	Ch 13
sort	group by a rule	Ch 7
swap	change places	Ch 10
trace	write the number after each step	Ch 12
variable	a named box for a number	Ch 12
view	how a thing looks from one side	Ch 2

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Cut-out code cards

Ask an adult to help you cut along the dotted lines. Use the cards to build programs on your desk, then act them out with a friend.

up	down	left
up	down	left
right	Start	Repeat ___ times
right	Start	Repeat ___ times
If ___	else	When ___ happens
If ___	else	When ___ happens
add ___	take away ___	Stop
add ___	take away ___	Stop

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